

16 Words (NYT Connections Puzzle)

Stage I — Data Preparation

Chronological Split
640 / 137 / 138 games

Data Augmentation
Synonym Substitution

Stage II — Word Representation

MPNet / E5-Large
768-dim

ConceptNet Numberbatch
300-dim

Joint Embedding (MPNet $\oplus$ Numberbatch, 1,068-dim, L2-normalised)

Stage III — Pairwise Similarity ($9 \times 16 \times 16$ matrices)

Semantic
MPNet cosine

Lexical
char n-gram

WordNet
Wu-Palmer

WikiData
Jaccard

Datamuse
Jaccard

ConceptNet
Jaccard

Wikipedia
Categories

Phonetic
CMU Dict.

Morphological
Affix/Lemma

Double-Centering Normalisation $\tilde{S}(i, j) = S(i, j) - \bar{S}_i - \bar{S}_j + \bar{S}$

Stage IV — Downstream Pipelines

Path A — Inference

Weighted Matrix Fusion
Learned weights

Beam Search Solver
 $K = 5$, two-step look-ahead

Uncertainty Filter
RF vs. GBT agreement

Path B — Training

Feature Extraction
~60 features per candidate

Ensemble Classifier
Random Forest + Grad. Boost

Beam Search Evaluation
 $K = 5$, partition check

4 Groups of 4 Words (Predicted Partition)